

Extrusion-based 3D-printable amyloid nanofibril-based hydrogel: A multifunctional system for hemostasis and localized drug delivery applications

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ARTICLE INFO

Keywords:

Curcumin
Amyloid nanostructures-based hydrogel
3D-bioprinting
Hemostasis
Drug delivery

ABSTRACT

Recently, amyloid nanostructures-based hydrogels are gaining prominence in biomedical research for their exceptional mechanical stability, biodegradability, and biocompatibility. Herein, we propose a novel 3D-printable amyloid nanostructures-based hydrogel that encapsulates curcumin into the fibrillar aggregates derived from bovine serum albumin (BSA), aiming to combine the inherent mechanical stability of fibril structures with curcumin's bioactive properties. The resulting composite hydrogel exhibits porous structure, well-regulated rheological characteristics with excellent injectability and 3D-printing properties. According to ABTS (2,2'-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid)) and 2,2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging tests, curcumin significantly enhanced the hydrogels' antioxidant activity. Furthermore, the in vitro drug release tests demonstrated a sustained release of curcumin by either swelling or diffusion, demonstrating the exceptional effectiveness of amyloid-based hydrogel as a drug delivery vehicle for curcumin. Biocompatibility and cytotoxicity of the hydrogels were evaluated through in vitro cytotoxicity assays, which confirmed no toxic effects on L929 mouse fibroblast cells, supporting the initial suitability of the hydrogel for cellular applications. Additionally, the resulting hydrogel possesses potent hemostatic capabilities, facilitating rapid blood coagulation. The amyloid nanostructures-based hydrogel emerged as a promising candidate for applications in hemostasis and localized therapeutic delivery.

1. Introduction

Amyloid nanostructures-based biomaterials have gained popularity and garnered significant interest recently because of their remarkable self-assembly properties, high mechanical strength, and stability in a variety of matrices [1,2]. Amyloid nanofibrils (AF) are insoluble protein aggregates characterized by β -sheet structure which are formed when soluble proteins misfold and aggregate [3,4]. The aggregates are formed

by heating the native protein solution which leads to misfolding and self-assembly of protein nanofibrils into a three-dimensional network. The formation of amyloid nanofibrillar aggregates starts by the growth of oligomers followed by the elongation into longer fibrils which can disrupt normal cellular functions [5,6] and lead to tissue damage which are associated with diseases like Alzheimer's, Parkinson's and type II diabetes etc [7,8]. Even though these amyloid aggregates are associated with various debilitating disorders, the highly ordered structures of

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<https://doi.org/10.1016/j.jddst.2026.108262>

Received 24 December 2025; Received in revised form 10 March 2026; Accepted 20 March 2026

Available online 21 March 2026

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amyloid nanostructures are nowadays considered as an ideal candidate for applications in tissue engineering and drug delivery [9,10] due to their varied structural properties as well as their excellent biodegradable and biocompatible nature [11,12]. These characteristics have expanded the possibilities for fabricating novel functional amyloid-based materials, including scaffolds [13,14] nanoparticles [15], membranes [16, 17] and hydrogels [18,19] etc.

Among the various amyloid-based materials, amyloid-based hydrogels have recently gained interest for 3D-printing purposes to design customized three-dimensional scaffolds for biomedical applications [20, 21]. Hydrogels are three-dimensional networks which have tunable physical properties and extracellular matrix mimicking properties. Along with that, hydrogels can be incorporated with various bioactive substances for enhanced tissue repair, drug delivery and healing purposes [22,23]. Amyloid nanofibrils can be assembled and easily form hydrogel material by physical interactions including electrostatic interactions, hydrogen bonding interactions, π - π interactions [24], and hydrophobic interactions [25]. These hydrogels also exhibit several unique physiognomies, such as high-water retention properties [26], higher aspect ratio [27], interface adsorption [28], good mechanical strength and biocompatibility [29,30]. Furthermore, the thixotropic and rheological properties make the hydrogels suitable for printing purposes and further use in clinical fields, especially for wound healing and tissue engineering applications [31]. In addition, 3D-printable hydrogels can be used as a skin-adhesive [32] with higher water absorption capabilities that help to maintain moisture at the wound site, which would promote healing of the wound [33]. Considering the importance of amyloid nanofibril-based hydrogels in clinical research, researchers have started focussing on fabricating novel amyloid-based hydrogels by combining β -lactoglobulin (β -LG) with bovine serum albumin (BSA) [34], nanoengineered DNA hydrogel through co-assembly of amyloid nanofibrils and clay nanosheets [35], and hen egg white lysozyme (HEWL) [36–38] etc. However, there are limited reports of 3D-printable amyloid-based hydrogels. Injectable and 3D-printable peptide hydrogels have been a topic of interest due to the shear induced alignment and templating of material into a predetermined structure [39–41]. Considering the characteristics of peptide hydrogels and its alignment with the conditions of biomaterial ink for 3D-printing [42,43], in our current work our major aim is to design 3D-printable amyloid nanostructure-based hydrogel using BSA as a model protein.

Curcumin is a hydrophobic compound which is characterized by phenolic groups and conjugated double bonds in its structure. It has wide range of bioactivities like anticancer, anti-inflammatory, antibacterial and antioxidative property [44,45] which makes it ideal to use in wound healing and tissue engineering applications [46–48]. Furthermore, curcumin has excellent reactive oxygen species (ROS) scavenging activity thereby promoting angiogenesis and collagen deposition in wound sites. Despite its enormous biological potential, curcumin's main drawback is its poor solubility in water, which also limits its use in the clinical area by causing poor bioavailability and instability [49,50]. In order to address these issues, researchers have focused on encapsulating curcumin into various nanoparticulate system such as self-assembled nanoparticles [51–53], extracellular vesicles like exosomes [54–56], niosomes [57,58], hydrogels [59–61] etc. Among the numerous matrices, amyloid -based hydrogels have been regarded as an exceptional platform because of their high surface hydrophobicity, excellent mechanical capabilities, high aspect ratio, and highly organized topologies [62].

Although curcumin loaded hydrogel systems are discussed extensively in the recent years, most of them are focussed on drug encapsulation and release properties while very few explores the chance of using curcumin in modifying the hydrogel matrix design and processability. To the best of our knowledge, no studies have been done on preparing an amyloid based curcumin loaded hydrogel which can be 3D printed and simultaneously uses curcumin as a rheology modifier apart from its hemostatic, antioxidant, and drug-delivery functionalities within a

single platform. Therefore, in the current study, we have synthesized a 3D-printable amyloid nanofibril-based hydrogel from BSA (Scheme 1) and further incorporating curcumin within it to investigate the therapeutic efficacies for further biological applications. This work introduces a novel 3D-printable hydrogel fabricated exclusively from BSA and curcumin, eliminating the need for external crosslinkers or synthetic additives. The hydrogel's structural integrity is achieved through self-assembled amyloid fibrils, while curcumin acts as a dual-functional component, enhancing both printability and bioactivity. Unlike conventional hydrogels, this system simplifies fabrication by avoiding harsh chemicals, offering improved biocompatibility. Even though the intervention of curcumin in the amyloid aggregation process is studied in molecular level, the rheological modification resulted from this interaction is not vastly utilised for printing applications. In our system, curcumin not only act as a therapeutic component but also as a structural modulator for BSA fibrillar network. The printability of the hydrogel can be easily tuned by adjusting the curcumin loading percentage. Moreover, curcumin shows increased solubility and stability inside the fibril which is beneficial for its controlled release. Considering the increasing importance of 3D bioprinting in tissue engineering and biomedical applications, we believe that the current research work will pave way to design novel 3D-printable amyloid -based hydrogels for applications in regenerative medicine and therapeutic delivery.

2. Experimental section

2.1. Materials

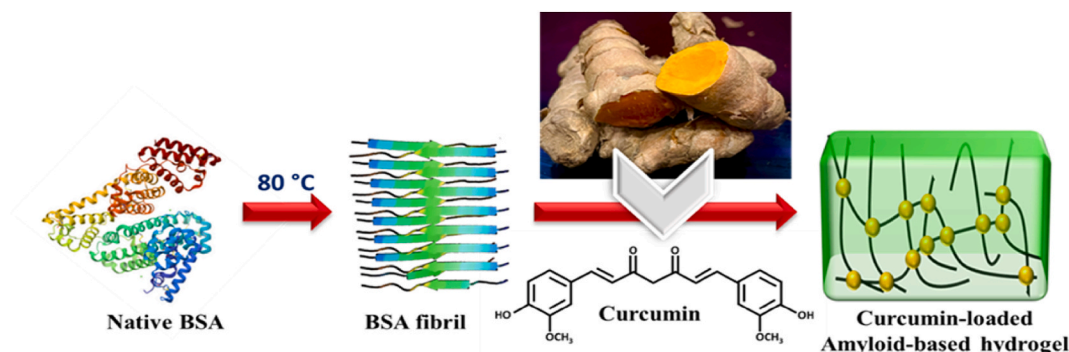
Bovine serum albumin (BSA), sodium chloride, sodium dihydrogen phosphate, disodium hydrogen phosphate, thioflavin T (ThT), ABTS (2,2'-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid)) and curcumin were procured from Sisco Research Laboratories Pvt. Ltd. (SRL), India and used as received. For the cell culture experiments, the L929 mouse fibroblast cell line was acquired from NCCS, Pune, India. MTT (3-(4,5-dimethylthiazol-2-yl) 2,5 diphenyltetrazolium bromide), dimethyl sulfoxide (DMSO), Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum were obtained from HIMEDIA (Mumbai, India). Cells were maintained at 37 °C in 5% CO₂ humidified atmosphere. The Live/Dead Cell Imaging Kit was procured from Invitrogen by Thermofischer Scientific. Milli-Q grade water was used throughout all the experimental work. All the chemicals used for our study were of analytical grade and were used without further purification.

2.2. Synthesis of amyloid-based hydrogel

At first, 0.1 g/mL concentration of BSA solution was prepared by dissolving native BSA in phosphate buffer saline (PBS) (pH = 7.2, 35 mmol NaCl). The BSA solution was then allowed to heat in a hot water bath by maintaining the temperature at 80 °C. After a period of 25 min, when slight turbidity starts to appear, curcumin solution in ethanol (1.2 mg/mL) was added to the BSA solution keeping the final concentration of curcumin as mentioned in Table 1. The heating was then continued for another 30 min until it passes the vial inversion test for the formation of a hydrogel. Amyloid hydrogel in absence of curcumin is denoted as **AF0** whereas curcumin-loaded amyloid-based hydrogels are denoted as **AF1**, **AF2** and **AF3** as shown in Table 1.

2.3. Printing of hydrogels

An extrusion bioprinter TRIVIMA with double extruder from NBIL (India) was used to fabricate the scaffolds. The hydrogels were loaded into a syringe with 26 gauge (279 μ m diameter). Scaffolds of dimensions 20x20x0.4 (mm) were printed at a speed of 50 mm min⁻¹ at 25 °C.



Scheme 1. Schematic representation of synthesis of curcumin-loaded amyloid nanostructure-based hydrogel.

Table 1

Sample composition for the synthesis of curcumin-loaded amyloid-based hydrogel.

Sample code	BSA (w/v %)	Curcumin (μM)
AF0	10	0
AF1	10	155 μM
AF2	10	296 μM
AF3	10	425 μM

2.4. Characterization of amyloid fibrils (Thioflavin T assay)

The formation of amyloid fibrils was confirmed by Thioflavin T (ThT) assay [63]. ThT is a fluorescent probe which shows an increase in intensity on conjugating with amyloid fibrils [64]. At first, 2.66 μL of aliquots from the amyloid fibril solution were taken and 10 μL of ThT (1 mM) solution was added to it. Then the mixture was diluted with PBS accordingly to keep the final concentration of samples and ThT at 2 μM and 5 μM respectively. After that, the solution was incubated for 5 min before recording the spectra. The fluorescence spectra were then measured in Hitachi Spectrofluorometer (F4700) in the range of 460 and 650 nm by exciting them at 450 nm. The other parameters like slit width and integration time were fixed at 5 nm and 0.1s respectively.

2.5. Vial inversion test

The formation of amyloid-based hydrogel was confirmed through vial inversion test. After the formation of amyloid fibril solution, the sample was kept at a fixed temperature of 80 $^{\circ}\text{C}$ and allowed for gelation. The gelation strength was then tested by keeping the vial inverted.

2.6. Morphological analysis

To evaluate the microstructural features such as surface morphology and interconnectivity of the lyophilized hydrogel, scanning electron microscopy (SEM) (Carl Zeiss Sigma) was used. The freeze-dried hydrogel sample was carefully mounted on a double-sided conductive carbon tape to ensure proper adhesion to the sample holder. Prior to imaging, the sample was subjected to gold sputter coating using a sputter coater to enhance electrical conductivity and prevent charging effects under the electron beam.

2.7. Fourier transform infrared (FTIR) spectroscopy

The FTIR spectra of samples were measured using a Shimadzu IRTracer-100 FTIR spectrophotometer equipped with an attenuated total reflectance (ATR) accessory. The background was composed of air. Spectra were recorded in the wavenumber range of 4000 to 400 cm^{-1} , with a resolution fixed at 4 cm^{-1} and averaged over 24 scans.

2.8. Injectability study

Injectability test provides an initial indication of the hydrogel's suitability for applications in 3D bioprinting, drug delivery, or minimally invasive therapeutic techniques [65]. Briefly, the hydrogels were carefully loaded into a syringe and extruded through a 21G needle to assess their extrudability. This process involved applying controlled manual or mechanical pressure to the syringe to observe the ease with which the hydrogel could pass through the narrow bore of the needle. The 21G needle, with an internal diameter of approximately 0.51 mm, was chosen to simulate extrusion conditions, often encountered in biomedical or engineering applications. The extrudability was qualitatively evaluated based on the uniformity and continuity of the extruded hydrogel filament, as well as the force required to initiate and maintain extrusion.

2.9. Rheological measurements

Rheological studies were carried out using MCR 302 rheometer from Anton Paar. The synthesized hydrogel was loaded between the parallel plates (diameter 25 mm) with a gap of 0.1 mm gap. The viscoelastic behaviour of the hydrogels was captured by conducting frequency sweep. More specifically, a range of frequency from 0.1 to 10 rad/s at a constant strain of 1% was imposed to measure the storage modulus (G') and loss modulus (G''). Furthermore, the shear thinning behaviour of hydrogels was studied by capturing the viscosity as a function of shear rate from 0.1 s^{-1} to 100 s^{-1} . Additionally, temperature sweep experiment was carried out over the range 25–55 $^{\circ}\text{C}$ using a strain of 1% at constant frequency of 10 rad/s to understand the effect of temperature on the mechanical stability of the synthesized hydrogel in terms of storage modulus (G') and loss modulus (G'').

2.10. Printability studies

The hydrogel scaffolds were fabricated using an extrusion-based bioprinter TRIVIMA equipped with a double extruder system (NBIL, India). For printing, the hydrogels were loaded into a standard syringe fitted with a 26-gauge nozzle (inner diameter: 279 μm). The printability tests include (a) printable angles, (b) printability factor determination and (c) filament width, smoothness, and d) fusion at crossover points. To check the printable angles, CAD design of a square was made and 3D printed [66]. The printing parameters were optimized to ensure precise deposition, with square scaffolds designed to dimensions of 20 mm (length) $\times$ 20 mm (width) $\times$ 0.4 mm (height). The extrusion was performed at a constant speed of 50 mm min^{-1} , and the printing process was carried out at room temperature (25 $^{\circ}\text{C}$) to maintain hydrogel stability and prevent premature gelation. This setup allowed for controlled layer-by-layer fabrication of the 3D structures while preserving the hydrogel's structural and functional properties. To check the printability factor, the constructs were printed, and then area (A) and perimeter (L)

were determined using Image J software and the printability factor (Pr) was calculated using Equation (1). Filament width, smoothness, and filament fusion at crossover points were also determined using Image J software.

$$Pr = \frac{L^2}{16A} \quad (1)$$

2.11. Swelling study and degradation studies

Swelling ability of the hydrogels were monitored by immersing a fixed amount of lyophilized samples in PBS (pH 7.4 and pH 5.5) for a period of 24 h. After the immersion time, the samples were taken out and the excess PBS was removed by gently tapping with a tissue paper and the weight was determined. The swelling ratio, S was calculated as follows:

$$S = \frac{m_t - m_0}{m_0} \quad (2)$$

Where m_t represents the mass of hydrogel at time t and m_0 represents the initial weight of the lyophilized hydrogel.

Furthermore, in vitro degradation behavior was assessed by measuring the percentage of mass loss over time by weighing a specific quantity of each lyophilized hydrogel samples and immersing it in a buffer solution (pH 5.5). After incubating for 7 days, the hydrogels were taken out, dried and weighed. The experiment was repeated in triplicates.

2.12. In vitro release study

In vitro release study was performed [67] to monitor the release of curcumin from the AF hydrogels. Briefly, 200 μ L of hydrogels were immersed in 5 mL of phosphate buffer (pH 5.5) containing equal amount of ethanol considering the poor solubility of curcumin in water. A PBS: ethanol (1:1v/v) is usually used in drug release studies of hydrophobic drug such as curcumin to ensure adequate drug solubility [68] apart from the encapsulation of curcumin in hydrophobic systems for preventing leaching. PBS provides adequate physiological environment while ethanol acts as a co-solvent to enhance the solubility of curcumin and maintain sink conditions throughout the release study. This release media can prevent curcumin precipitation, enables accurate and reproducible quantification of released drug [69,70]. The release media were kept shaking at 100 rpm. At specific time intervals, 2 mL of aliquots were drawn at specific intervals and immediately replaced with fresh medium to maintain total volume. The absorbance of the solution was taken at 425 nm using UV-Vis Spectrophotometer (Multiskan SkyHigh Microplate Spectrophotometer (A51119500C)).

2.13. Antioxidant studies

a) **DPPH scavenging assay:** 2,2-diphenyl-1-picrylhydrazyl (DPPH) assay [71] was performed to determine the antioxidant activity of the AF hydrogels in absence and presence of curcumin. The stock solution of DPPH (1 mM) was made in ethanol. 50 mg each of the hydrogels were immersed in 3 mL of DPPH solution and kept in dark for a period of 40 min. After the incubation period, the absorbance of the solution was measured at 517 nm using a UV-Vis spectrophotometer (Multiskan SkyHigh Microplate Spectrophotometer (A51119500C)). The DPPH radical scavenging activity was then calculated using equation (3):

$$DPPH\text{ Scavenging activity} = \frac{A_{blank} - A_{sample}}{A_{blank}} \times 100 \% \quad (3)$$

A_{blank} is the absorbance of the DPPH solution and A_{sample} is the absorbance of the hydrogel immersed in DPPH solution. Each sample was tested in triplicate for reproducibility of the study.

b) **ABTS scavenging assay:** ABTS scavenging assay was carried out

according to Sen et al. [72] with slight modifications. ABTS radical cation solution were generated by combining a 7 mM ABTS solution with 2.4 mM potassium persulfate in a total volume of 5 mL, and this mixture was allowed to incubate in the dark for 14-16 h. Subsequently, the working ABTS solution was diluted with PBS (pH 7.4) to achieve an absorbance of 0.7 ± 0.02 at 734 nm. In 3 mL of this freshly prepared ABTS working solution, 100 μ L of each hydrogel was added, and the absorbance was recorded after 40 min. The radical scavenging activity was determined in a manner similar to the DPPH method.

2.14. Hemolysis assay

Hemolysis assay [73] was conducted to determine the hemolytic activity of the synthesized hydrogels. At first, the collected blood was centrifuged at 4500 rpm for 15 min and continuously washed with PBS (pH 7.4) to separate the red blood cells (RBCs) from plasma and buffy coat. 50 mg of hydrogel samples were then incubated with RBC suspensions (1% hematocrit) for 40 min. After the incubation period, the samples were again centrifuged at 4500 rpm for 10 min. The absorbance of the supernatant was taken at 540 nm using UV-Vis spectrophotometer (Multiskan SkyHigh Microplate Spectrophotometer (A51119500C)). In this study, RBCs in water was taken as positive control and RBCs in PBS was taken as the negative control. The hemolysis percentage was measured by using the following equation (4):

$$\% \text{ Hemolysis} = \frac{A_{sample} - A_{negative\ control}}{A_{positive\ control} - A_{negative\ control}} \times 100 \quad (4)$$

Where A_{sample} is the absorbance of the sample solution, $A_{positive\ control}$ is the absorbance of the positive control (RBCs in water) and $A_{negative\ control}$ is the absorbance of the negative control (RBCs in PBS).

2.15. Measurement of activated partial thromboplastin time (APTT) and prothrombin time (PT)

Automatic coagulation analyzer was used to determine the coagulation parameters such as partial thromboplastin time (APTT) and prothrombin time (PT). In order to detect the parameters, the collected blood was centrifuged at 4500 rpm for 20 min to obtain the test plasma.

2.16. In vitro cytocompatibility studies

All four cell-seeded hydrogels after culturing for 24 h and 48 h were analyzed through 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay. Briefly, 5 mg/mL concentration of MTT reagent was added to the media and incubated at 37 °C in 5% CO₂ for a period of 4 h. The media containing MTT was discarded after the incubation period followed by the addition of dimethyl sulfoxide (DMSO) to dissolve the purple formazan crystals completely into the solution. The absorbance at 570 nm was then taken using a microplate reader spectrophotometer. The optical density values give information about cell compatibility and viability. The cells grown in tissue culture plates were taken as positive control and the cells grown in blank tissue culture plates were taken as negative control. The experiment was done in triplicates for the reproducibility of the work. Finally, the % cytotoxicity and % viability were calculated using equation (5) and equation (6) respectively:

$$\% \text{ cytotoxicity} = \frac{A_{control} - A_{test}}{A_{control}} \times 100 \quad (5)$$

$$\% \text{ viability} = (100 - \% \text{ cytotoxicity}) \quad (6)$$

Where A_{test} represents the absorbance of the sample and $A_{control}$ signifies the absorbance of the control sample.

2.17. Live/dead staining viability assay

To further observe cellular proliferation on the hydrogels, the nuclei of cells attached to the hydrogels were observed using live/dead staining viability assay. For the live/dead staining assay, cells were seeded at a density of 20,000 cells per well in a multi-well plate. Following seeding, the cells were incubated with the hydrogel samples for 48 h under standard culture conditions. After the incubation period, the cells were gently washed with phosphate-buffered saline (PBS) to remove any residual medium or unattached cells. A staining solution was prepared according to the manufacturer's instructions using the reagents provided in the live/dead assay kit. This solution was then introduced into the wells containing the cells, and the plate was incubated in the dark for 20 min to ensure proper staining. After incubation, the cells were washed thoroughly with PBS to remove any unbound dye. Finally, the stained cells were visualized through fluorescent microscope. Images were then captured using both green and red fluorescence filters to differentiate between live cells (green) and dead cells (red), providing a clear representation of cell viability after exposure to the hydrogel samples.

2.18. Statistical analysis

All experiments were performed in triplicate unless otherwise specified. A value of $p < 0.05$ was considered statistically significant.

3. Results and discussions

Prior to the formation of curcumin-encapsulated amyloid fibril-based hydrogel, the formation of bovine serum albumin (BSA) amyloid fibrils was confirmed by Thioflavin T (ThT) fluorescence assay. ThT, a specific dye for amyloid fibrils, exhibits weak fluorescence in its free state but undergoes a significant increase in fluorescence intensity when bound to amyloid fibrils. Upon binding, ThT emits fluorescence with a maximum peak at 482 nm when excited at 450 nm, making it a highly sensitive marker for amyloid aggregation [74]. In our case, when the amyloid fibril precursor solution was treated with ThT, a significant increase in fluorescence intensity was observed around 482 nm as shown in Fig. 1A. This increase in fluorescence is a characteristic indicator of BSA amyloid fibril formation, as ThT is known to selectively bind to the β -sheet structures present in amyloid fibrils. Next, the morphology of BSA fibrils was monitored using transmission electron microscopy (TEM) and the image (the inset TEM image in Fig. 1A) clearly shows gives elongated, hair-like fibrillar structures which indicates the formation of BSA amyloid fibrils. Furthermore, Far-UV CD spectroscopic measurements were carried out (Fig. 1B) to identify the changes in

secondary structures of BSA protein after the formation of fibrils. As shown in Fig. 1B, native BSA exhibited two strong negative minima at ~ 208 and ~ 222 nm, which indicates a predominant α -helical conformation. In contrast, BSA fibrils showed a noticeable reduction in α -helical signals in addition to the appearance of a broad negative band around ~ 218 nm, indicating the formation of β -sheet-rich structures. The loss of α -helicity and emergence of β -sheet rich structures confirm that fibrillation involves partial unfolding of BSA followed by structural reorganization into an amyloid-like architecture.

After the successful synthesis of BSA amyloid fibrils, the curcumin-encapsulated amyloid fibril-based hydrogel was synthesized by adding curcumin into the fibrillar solution and heating the precursor solution to a temperature of approximately 80°C . This heating process facilitated the denaturation of BSA, promoting the self-assembly of amyloid fibrils. BSA contains two tryptophan residues one of which is in the hydrophobic pocket that can interact with ligands. The tryptophan residues give intense fluorescence emission when excited at 295 nm which is very sensitive to the local environment, intermolecular interactions and conformational alterations. Therefore, monitoring intrinsic tryptophan fluorescence can be indicative of the binding interactions between the ligand and BSA. Furthermore, curcumin is known to interact with BSA amyloid fibrillar structures through secondary non-covalent interactions such as H-bonding, π - π interactions, hydrophobic interactions etc which interferes with the fibril formation by destabilizing the β -sheet structures. Additionally, previous literature reports showed that molecular simulation and docking studies were used to understand the protein binding interaction between the closest molecular residues and the curcumin molecule. The theoretical studies confirm that the curcumin is held tightly in the binding site through mainly hydrophobic and hydrogen bond interactions [75,76]. This interaction between curcumin and amyloid fibrils can inhibit the fibril aggregation process to a certain extent which can be beneficial in making it 3D printable through careful tuning of curcumin concentration. When curcumin is added during the fibrillation stage, it interacts with the exposed tryptophan residues of amyloids and a significant dose-dependent quenching of intrinsic fluorescence emission of tryptophan of the amyloid aggregate system was observed (Fig. S1A). This indicates that curcumin can interact with the soluble intermediates-the precursor of amyloid fibrils, thereby altering the final fibrillar structure. This supports the existing studies about the ability of curcumin to prevent the exposure of hydrophobic residues and pack the protein in the native stage [77]. Furthermore, fluorescence properties of curcumin itself confirm its interaction with amyloid fibrils. Even though curcumin shows very weak intrinsic fluorescence [78], however, adding fibrillary solutions increases its fluorescence intensity and induces a distinct blue shift of ~ 50 nm (Fig. S1B). Due to the hydrophobic nature of curcumin, it can enter the hydrophobic core of

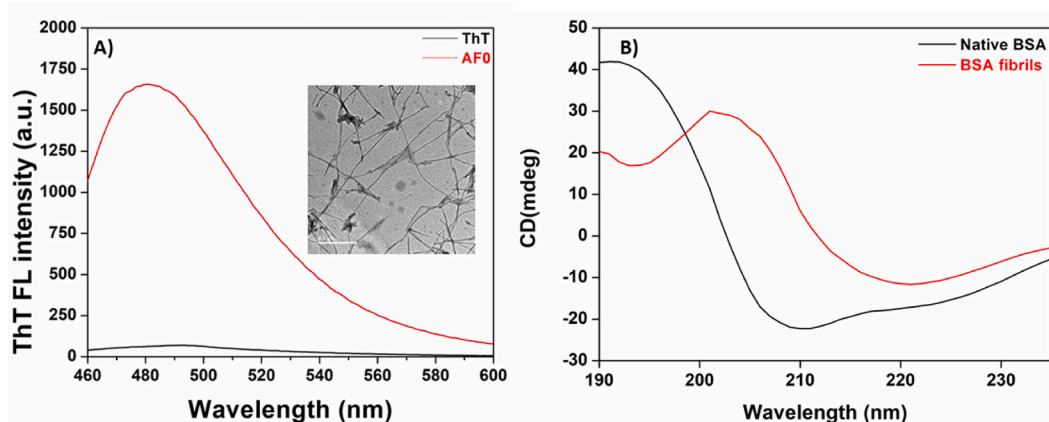


Fig. 1. A) Thioflavin T fluorescence spectra of amyloid fibrillar solution of BSA. The inset shows the hair like fibrillar morphology of BSA amyloids B) Far UV-CD spectra of native BSA and BSA fibrils.

fibrils and undergo self-aggregation at the fibrillar surface in aqueous environment [79].

After encapsulating curcumin into the fibrillary networks, the solution was heated until it transformed into a translucent gel, with successful gelation. The stability and mechanical integrity of the gel were validated using the vial inversion test, where the gel retained its structure without flowing upon inversion of the container. As shown in vial inversion test (Fig. 2A), initially, the BSA solution exists in a clear solution phase, however, it transitioned into a gel-like state upon heating indicating the formation of amyloid fibril-based hydrogel (Fig. 2B). This change in physical consistency, from a free-flowing liquid to a solid state was a direct consequence of the self-assembly of BSA into amyloid fibrils and further forming hydrogel under heat-induced stress. The resistance to flow observed during the vial inversion test provide strong evidence for the presence of these amyloid-based hydrogel.

Fourier-transform infrared (FTIR) spectroscopic measurement was carried out to detect the functional moieties present in the synthesized hydrogels. As shown in Fig. 3A, the characteristic amide peaks at approximately 1530 cm^{-1} and 1645 cm^{-1} confirm that the protein backbone remains intact across all composite formulations, indicating the preservation of the structural integrity of the BSA within the hydrogels. A decrease in the intensity of the amide bands in curcumin-loaded amyloid-based hydrogel may be due to the formation of hydrogen bonds between curcumin and BSA. This interaction suggests a significant molecular interaction that could influence the mechanical and functional properties of the hydrogel. Additionally, as the concentration of curcumin in the hydrogel increases, the peak observed around 3300 cm^{-1} , associated with the N-H stretching vibrations, exhibits a weakening trend. This reduction in peak intensity further supports the hypothesis that curcumin interacts with BSA through hydrogen bonding, leading to a disruption of the N-H bond characteristics. Such changes in the FTIR spectra shows the influence of curcumin to modify the biochemical environment within the hydrogel matrix, which can help for its further targeted applications.

The injectable properties of the synthesized hydrogels were monitored by performing injectable study. Interestingly, the injectability test of the hydrogels demonstrated that both AF2 and AF3 can be smoothly extruded through a 21G needle without any structural breakage. However, the injectability of AF1 is not smooth, leading to inconsistent flow while trying to print with a needle. This irregularity causes the material to break down or disrupt the intended structure, resulting in incomplete structures. The lack of smooth extrusion reduces the precision and stability of the printed scaffolds. On the other hand, in case of AF2 and AF3, no evidence of clogging or gel disruption was observed during the process, indicating that both AF2 and AF3 hydrogels maintain their integrity under the mechanical shear forces during injection. These findings suggest that these hydrogels possess optimal rheological characteristics which can be helpful for bioprinting. These results further indicate that the incorporation of curcumin significantly enhanced the

injectability of the hydrogels facilitating smoother extrusion through the needle without affecting the structural integrity. This enhancement in injectability suggests that curcumin not only contributes to the hydrogel's mechanical stability but also optimizes its flow behaviour, making it more suitable for precise, controlled delivery in clinical applications. The injectability of the AF hydrogels is shown in Fig. 3B.

Furthermore, to study the viscoelastic properties of synthesized hydrogels, rheological analysis was conducted. An ideal biomaterial ink for extrusion-based 3D bioprinting should exhibit shear-thinning property which is necessary for ensuring smooth extrusion through the nozzle [80]. A frequency sweep measurement was conducted within the LVR to characterize the structural and viscoelastic properties of the hydrogels. During the sweep, a constant strain of 1% was applied, and the storage modulus (G') and loss modulus (G'') were plotted against the frequency. As depicted in Fig. 4A, all hydrogels exhibit $G' > G''$ across the frequency range, highlighting their predominantly elastic nature and long-term stability due to a stable network of interactions. Additionally, both G' and G'' remain nearly constant and frequency-independent across the entire range from 0.1 to 10 rad/s. Generally, shear-thinning behaviour is defined by a reduction in viscosity as the shear rate increases, which leads to an easier flow during the printing process. Interestingly, AF2 and AF3 demonstrated excellent shear-thinning behaviour across the tested range of shear rates (0.1 to 100 s^{-1}) (Fig. 4B). Specifically, the viscosity of the hydrogels decreased significantly as the shear rate increased, confirming that the curcumin-loaded formulations are suitable for extrusion-based bioprinting. Comparing to AF0 and AF1 which shows a relatively abrupt decrease in viscosity after 10 s^{-1} , AF2 and AF3 has a smooth curve which shows a slow decrease of viscosity that is beneficial to the printing process. This slow and smooth reduction in viscosity comes under the requirements for bioprinting, where the material needs to transition from a more viscous state at low shear rates to a fluid-like state under the higher shear conditions experienced during extrusion. At shear rates below 1 s^{-1} , the hydrogels maintain high viscosity, indicating sufficient structural integrity for handling. However, as the shear rate increases beyond this threshold, the viscosity begins to decrease rapidly, reaching values within the optimal range for extrusion.

Temperature sweep measurements of the hydrogels were carried out to understand its thermal stability over a wide range of temperature. Interestingly, all the synthesized hydrogels show a dominant storage modulus (G'), indicating a predominant solid, gel-like structure at lower temperatures (Fig. 4C). The presence of curcumin likely strengthens the hydrogel's network, either by interacting with the amyloid fibrils or through other molecular interactions, making the material more resistant to temperature-induced structural changes. As a result, the hydrogel maintains its gel-like properties over a wider range of temperatures, preventing premature transitions to a more fluid-like state. The curcumin-loaded hydrogels therefore exhibit the necessary rheological properties for successful application in 3D bioprinting.

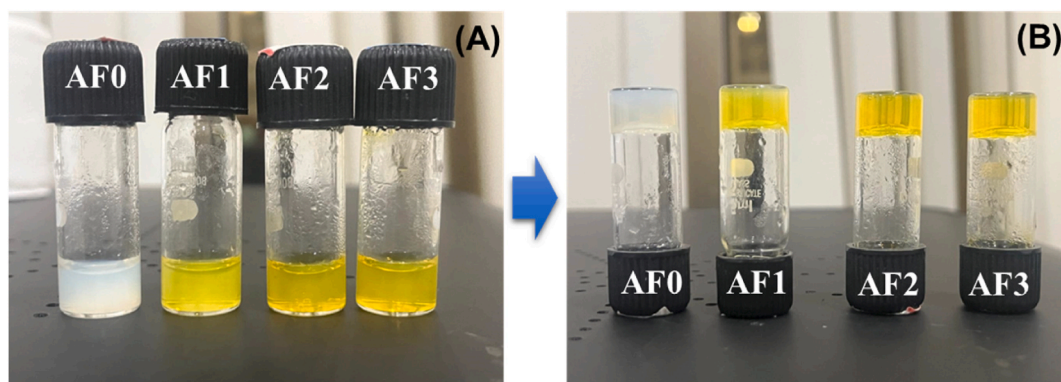


Fig. 2. Vial inversion test (A) before gelation and (B) after gelation.

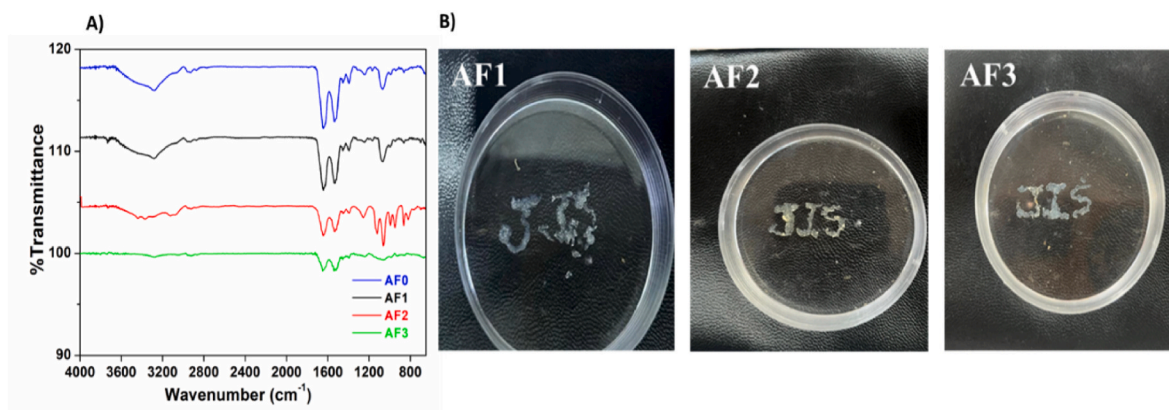


Fig. 3. A) FTIR spectra of amyloid-based hydrogels in absence and presence of curcumin B) Injectability test of AF hydrogels.

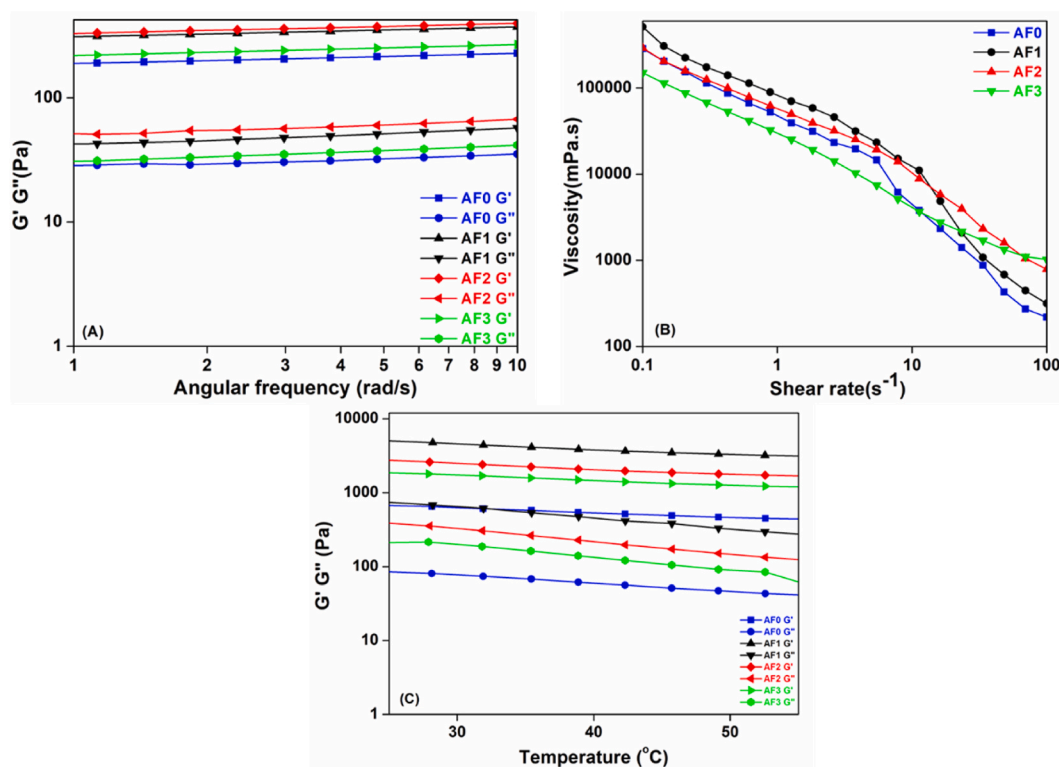


Fig. 4. Rheological characterization of AF hydrogels; (A) Frequency sweep at a constant shear strain of 1% and (B) Viscosity Vs shear rate curve (C) Temperature sweep measurements at constant frequency 10 rad/s and strain 1 %.

The printability studies are helpful for understanding the efficiency of the hydrogels for the 3D printing process. The details of printability study of the hydrogels were provided in Table 2. Interestingly, the results revealed that AF2 has a printability factor of 1.026, which is slightly above 1. This indicates good printability, as values around 1 are often considered optimal for extrusion-based 3D printing [81]. On the other hand, AF3 shows a printability factor of 0.997, which is slightly

lower but still close to 1. AF2 was printed at an angle of 93.26° at the corner, while AF3 was printed at 91.636°. Both angles are close to 90°, which is typical for a well-controlled, perpendicular filament deposition [82]. Moreover, AF2 has a filament width of 0.67 mm, while AF3 has a width of 0.61 mm. Generally, narrower filaments give higher resolution, a slight variation in filament width can affect the overall precision of the printed object. AF2 has a junction width of 0.81 mm, and AF3 has a junction width of 0.90 mm suggesting that both samples exhibit similar levels of fusion at filament junctions. It is also important to note that both AF2 and AF3 appears to have good printability based on the printability factor and narrower filament width, which could result in higher-resolution printing. Also, the printed structures exhibited good shape fidelity and structural stability during and after printing. The scaffold retained its predetermined structure without visible shrinkage and deformation. The differences between AF2 and AF3 are minor, and both show good potential for hydrogel printing. The printed constructs

Table 2
Printability study of hydrogels.

Sample code	Printability factor (Pr)	Printing angle (°)	Filament width (mm)	Filament fusion at the crossover point. (Junction width) (mm)
AF2	1.026	93.26	0.674	0.815
AF3	0.997	91.636	0.610	0.909

of hydrogels are shown in Fig. 5.

As mentioned earlier, curcumin interacts with BSA fibrils mainly through non-covalent interactions which loosens the network from the tight and dense fibrils of pure BSA. Curcumin interacts with the hydrophobic binding pockets of the protein and acts like an additional physical crosslinking factor imparting shear thinning property to the fibril hydrogel and excellent shape fidelity after printing. Additionally, the concentration of curcumin highly influences the printability and performance of the hydrogels. In our hydrogel system, pure BSA fibril hydrogel was unable to print because of its stiff structure contributed by the strong fibril-fibril interaction. As curcumin is introduced in the fibril system, the network started to be soft and in a moderate curcumin loading concentration, the hydrogel became injectable and printable with good structural stability.

The surface morphology and porosity of the freeze-dried printed scaffolds of AF2 and AF3 were then examined using scanning electron microscopy (SEM), as illustrated in Fig. S2. The scaffold of AF2 demonstrated a highly porous, honeycomb-like architecture, characterized by well-defined interconnected pores that facilitate nutrient diffusion and cellular infiltration. This distinctive morphology is conducive to tissue engineering applications, providing an optimal environment for cell attachment and growth. The scaffold of AF3 showed relatively dense, compact and rough structure indicative of high crosslinking density. This surface morphology suggests that AF2 and AF3 can provide support for cell proliferation and tissue integration.

The swelling behaviour of curcumin-loaded amyloid-based hydrogels were also monitored at pH 5.5 (Fig. S3A) and pH 7.4 (Fig. S3B) to check its effectiveness as a wound dressing material. The results revealed that at pH 7.4, AF2 showed the highest degree of swelling whereas AF0 exhibited a swelling percentage of 150%. This relatively low swelling capacity is contributed by the intrinsic properties of the amyloid fibril having limited ability to absorb water or expand. Upon the incorporation of curcumin into the hydrogel matrix, a notable enhancement in swelling behaviour was observed. In particular, the swelling percentage increased significantly in AF1, reaching 500%. The swelling percentage further increased to 550% in AF2, indicating that this formulation achieved the highest degree of swelling among the hydrogels tested. The increase in swelling in AF1 and AF2 can be due to optimal interactions between curcumin molecules and the amyloid fibrils, which allows further network expansion and create additional water-binding sites within the hydrogel. However, in AF3, a sharp reduction in swelling was observed, with the percentage decreasing to 175%. This decline in swelling capacity on increasing the curcumin loading may be due to the concentration dependent interactions between BSA and curcumin. At high concentrations, curcumin increase the

fibril-fibril interaction by acting as a physical crosslinker leading to a denser and tighter network which could lead to saturation of the binding site and lowering of porosity. Furthermore, at pH 5.5 also, it was observed that AF2 showed higher swelling with a swelling percentage of 364% followed by AF3, AF1 and AF0 with 328%, 270% and 235% respectively with all the hydrogel combinations having a swelling percentage above 200%. A higher swelling ability represents the hydrophilic nature of the hydrogel combinations which supports cell growth and nutrient transport. The swelling studies reveal that moderate curcumin incorporation (as seen in AF1 and AF2) enhances the hydrogel's swelling capacity, excessive loading (AF3) can lead to diminished swelling, likely due to changes in the structural integrity of the fibrillar network.

The in vitro degradation behaviour of the hydrogels were assessed over a period of 7 days at 37 °C at pH 5.5. After 7 days period, all the hydrogels showed a degradation rate above 80% with AF0 (84.8%), AF1 (87.4%) AF2 (92.3%) AF3 (85.5%) respectively which can be due to the predominant non-covalent interaction which facilitates hydrolytic degradation. It suggests that the hydrogels are biodegradable and are suitable to be used in physiological applications without long-term accumulation.

The DPPH and ABTS radical scavenging assays were performed to monitor the antioxidant properties of the curcumin-loaded amyloid fibril (AF) hydrogels. Curcumin is known for its antioxidant property due to its intrinsic polyphenolic structure, which enables efficient free-radical scavenging through hydrogen donation and electron transfer [83]. When incorporated into the hydrogel, non-covalent interactions with amyloid fibrils stabilize curcumin in a dispersed, non-aggregated state, preventing degradation and preserving its antioxidant functionality [84]. Interestingly, all the synthesized hydrogels exhibited notable radical scavenging capabilities in both the assays (Fig. S3C). The AF0 hydrogel, which does not contain curcumin, displayed a baseline DPPH RSA of 26.42% and ABTS RSA of 63%. However, upon the incorporation of curcumin, the radical scavenging activity of the hydrogels increased significantly. For the curcumin-loaded hydrogels, AF1 exhibited a DPPH radical scavenging activity of 46.52% and ABTS RSA of 72%, while AF2 and AF3 demonstrated 52.66% and 46.52% DPPH RSA respectively and 73.5% and 67.09% ABTS RSA, respectively. The enhanced RSA observed in these formulations is because of the presence of curcumin, which has significant antioxidant properties. The variation in RSA among AF1, AF2, and AF3 could be due to the differences in curcumin loading or interactions between curcumin and the amyloid fibril matrix, influencing the availability of curcumin's active sites for free radical neutralization. Notably, AF2 showed DPPH RSA of 52.66% and an ABTS RSA of 73.5%, suggesting that a higher availability of curcumin within

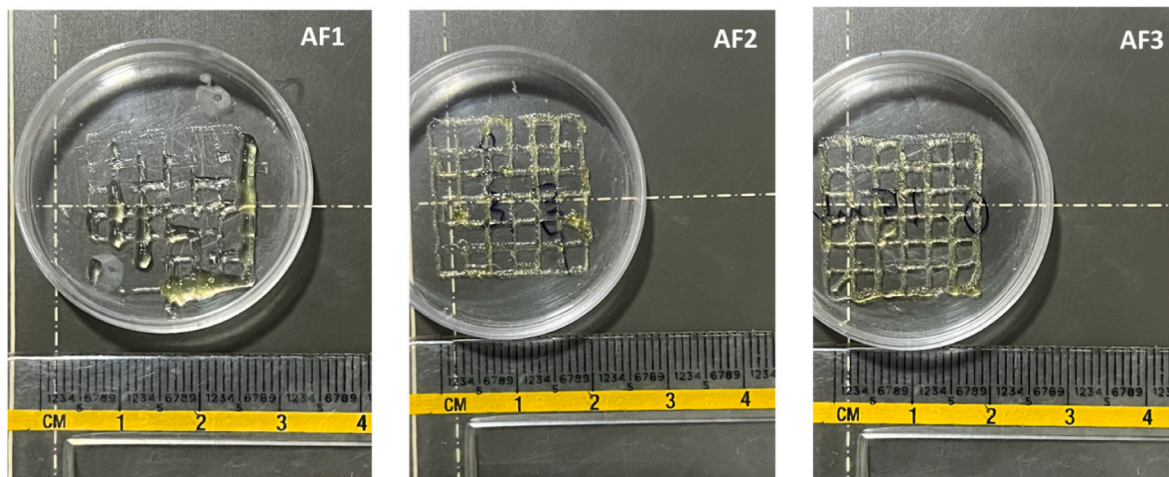


Fig. 5. Printability of AF hydrogels.

this formulation may contribute to its superior antioxidant performance.

The in vitro drug release study of curcumin-loaded AF hydrogels was carried out to evaluate the potential of amyloid-based hydrogels for controlled drug delivery. Drug release from hydrogel matrices typically occurs through a combination of desorption, diffusion, swelling, and polymer erosion, with diffusion and matrix degradation playing the predominant roles in modulating the release profile. As illustrated in Fig. S3D, the release of curcumin from all AF hydrogels exhibited a characteristic biphasic pattern, consisting of an initial burst release followed by a sustained release phase.

During the initial 1 h, AF1 exhibited ~21% release, AF2 ~41%, and AF3 ~24%, which can be attributed to the desorption of curcumin molecules adsorbed on or near the hydrogel surface. This burst release phase is beneficial for rapidly achieving therapeutic drug concentrations at the target site, which is particularly important in applications such as wound healing, where immediate action is required. Following this, the hydrogels entered the sustained release phase, during which curcumin embedded deeper within the hydrogel network was gradually released. This slow release occurs via diffusion through the hydrogel matrix and/or matrix relaxation and biodegradation, ensuring prolonged exposure of the drug at the site of action. At the end of 7 h, the cumulative release of curcumin from AF2 (~57%) was higher than AF3 (~33%) and AF1 (~29%), highlighting the influence of hydrogel network structure, porosity, and polymer chain mobility on release kinetics. The higher release from AF2 indicates a more flexible and hydrated network, which facilitates water penetration, matrix swelling, and enhanced diffusion of curcumin molecules. The reduction in drug release percentage in AF3 can be attributed to the stronger curcumin fibril interaction which stabilizes curcumin in the hydrophobic pockets of the fibril causing a comparatively slower release. The sustained release behaviour of AF2 qualifies its suitability for localized and controlled delivery, such as in wound sites, where maintaining therapeutic drug levels over an extended period is critical for effective healing and minimizing side effects. Moreover, the drug release study coheres with the radical scavenging activity of the hydrogels contributed mainly by the antioxidant properties of curcumin. Curcumin contains phenyl hydroxyl, β diketone, o-methoxy groups which can neutralize free radicals through mechanisms including hydrogen atom transfer, electron transfer. In this system the antioxidant activity is directly dependent on the amount of curcumin released because of its ability to stabilize the reactive oxygen species.

Considering the complexity of drug release process from hydrophilic hydrogel matrices, several mathematical models are used for understanding the release kinetics. In this study, the data were fitted into two main mathematical models i) Higuchi model (cumulative % drug released versus square root of time) and ii) Korsmeyer-Peppas model (log cumulative % drug released versus log time).

The data were first fitted into Higuchi square root kinetic model (cumulative % drug released versus square root of time) (Equation (7))

$$F = K_2 \times t^{\frac{1}{2}} \quad (7)$$

Where F signifies the cumulative percentage of drug released in time t and K_2 represents the Higuchi dissolution constant.

The Higuchi model plot (Fig. S4A) shows a linear relationship with a high R^2 value (0.933 for AF1, 0.923 for AF2 and 0.931 for AF3) which substantiate the diffusion-based release mechanism for curcumin from the hydrogels.

Furthermore, the data were fitted into Korsmeyer-Peppas kinetic model (log cumulative % drug released versus log time) (Equation (8))

$$F = K_1 \times t^n \quad (8)$$

where F represents the cumulative percentage of drug released in time t , K_1 is the release constant and n is the diffusional constant.

In Korsmeyer-Peppas kinetic model, $n \leq 0.45$, indicates Fickian diffusion mechanism (Case I diffusion). If the value of n comes between

0.45 and 0.89, it signifies non-Fickian type (Case II diffusion) of release mechanism (both diffusion and erosion). In our case, as represented in the Korsmeyer-Peppas model plot (Fig. S4B), curcumin release from AF1 exhibited a primarily Fickian diffusion with an n value of 0.368. AF2 is having an n value of 0.74 and AF3 is having an n value of 0.716 showed non-Fickian or anomalous transport, where drug release is controlled by a combination of diffusion through the polymer network and relaxation or swelling of the hydrogel matrix. These differences in release behavior can be attributed to variations in hydrogel microstructure, porosity, and crosslinking density, which influence both water uptake and polymer chain mobility, thereby altering the rate and mechanism of curcumin release. AF1 having diffusion controlled drug release may be suitable for applications requiring sustained, gradual drug delivery, whereas AF2 and AF3, with their mixed diffusion-relaxation behavior, could provide faster or more responsive release profiles.

Furthermore, hemolytic assay was performed to ensure the safety and biocompatibility of the synthesized hydrogels as it is essential to assess the toxicity of therapeutic compounds. According to ISO/TR 7405-1984(f) standards, materials are classified based on their hemolytic activity: those with less than 5% hemolysis are considered non-hemolytic, values between 5% and 10% are categorized as mildly hemolytic, and those with hemolysis rates exceeding 10% are considered highly hemolytic. In this study, the hemolytic activity of the curcumin-loaded amyloid fibril-based hydrogels was found to be below 2% (Fig. 6A). This indicates that the hydrogels are non-hemolytic, with no significant damage to red blood cells (RBCs) observed, thus demonstrating excellent hemocompatibility.

Next, in order to monitor the role of amyloid-based hydrogel as a hemostatic material, activated partial thromboplastin time (APTT) and prothrombin time (PT) were evaluated. The APTT tests (Fig. 6B) were performed to monitor the intrinsic pathway of thrombosis. When amyloid-based hydrogel without curcumin (AF0) is added to the blood, the APTT results showed an increase in APTT value (41.0 s) compared to the control (33.5 s) whereas the addition of curcumin results in decrease of APTT value of AF1 (37.1 s), AF2 (34.8 s) and AF3 (33.8 s) compared to AF0. This indicates that addition of curcumin into the amyloid-based hydrogel increases the coagulable ability of blood. Furthermore, PT tests (Fig. 6C) were carried out in order to evaluate the extrinsic pathway of thrombosis. The PT test results showed that even though AF0 (15.4 s), AF1 (14.8 s), AF2 (14.5 s) and AF3 (14.3 s) prolonged the PT compared to the control (12.8 s) but not significantly. From the above findings, we conclude that curcumin-loaded amyloid-based hydrogel acts as hemostatic material and the coagulation ability of curcumin-loaded amyloid-based hydrogel mainly occurs through intrinsic pathway of thrombosis. However, APTT and PT assays are considered as indirect evidence for the hemostatic potential of the material as they consider only the common coagulation pathways in controlled in vitro conditions rather than the actual in vivo hemostasis conditions. They only give the idea about the influence of material in hemostasis not about the exact hemostatic efficacy, platelet adhesion etc.

The MTT assay was conducted to evaluate the cytotoxic effect of curcumin-loaded amyloid-based hydrogels on the L929 cell line. At 24 h, the control sample had a mean O.D. of 0.6204 ± 0.0041 . Sample AF0 showed slightly lower O.D. values, with a decrease from 0.60955 ± 0.00475 at 5 mg/mL to 0.6009 ± 0.0007 at 10 mg/mL. Sample AF1 had a mean O.D. close to the control at 5 mg/mL (0.6125 ± 0.0138), but values decreased slightly with higher concentrations. Samples AF2 and AF3 followed a similar trend, with O.D. values decreasing as concentrations increased, with AF2 ranging from 0.5997 to 0.59325 and AF3 from 0.5984 to 0.58925. Overall, after 24 h of treatment, cell viability was above 90% for all the samples as compared to control, thus pointing to the cytocompatibility of the synthesized material (Fig. 6D). At 48 h, the control O.D. increased to 1.05075 ± 0.06575 . Sample AF0 exhibited high cell viability, with O.D. values ranging from 1.051 ± 0.028 to 1.0735 ± 0.0285 at 5 mg/ml and 10 mg/ml concentration respectively, suggesting good biocompatibility.

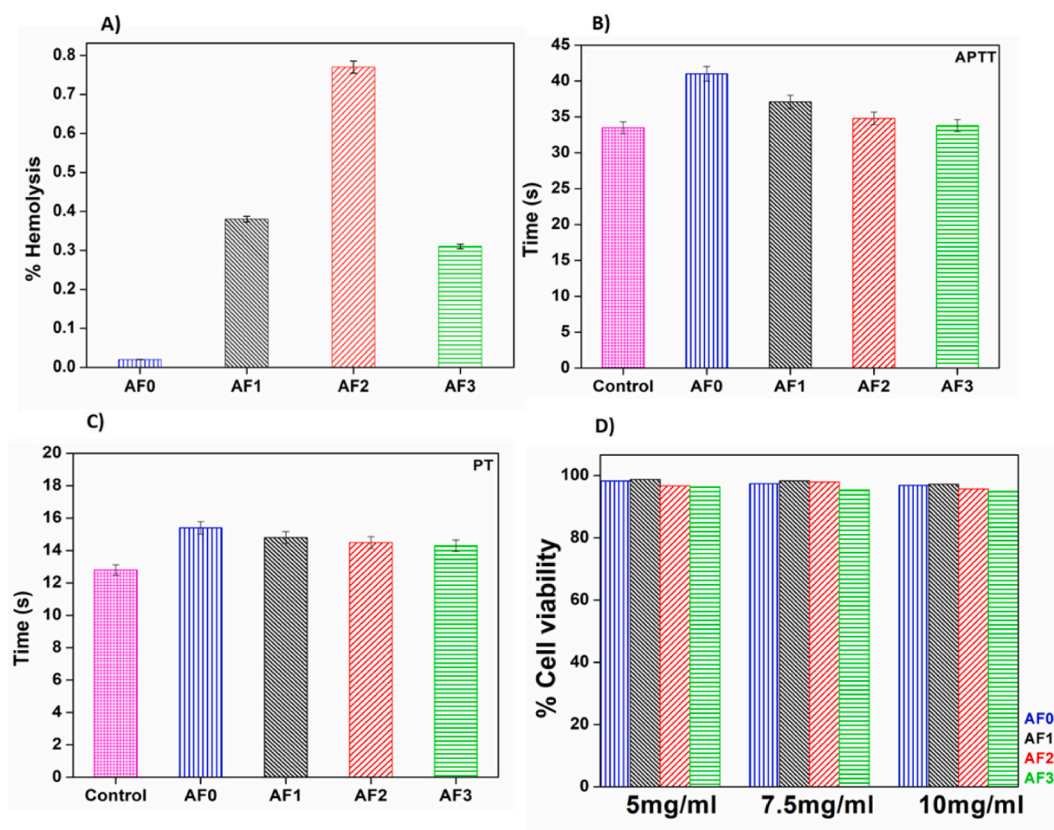


Fig. 6. (A) Hemolytic percentage of amyloid-based hydrogels in absence and presence of curcumin B) APTT tests and C) PT tests of AF hydrogels (D) Percentage viability on the L929 cell line after 24 h.

Sample **AF1** also showed high viability with values close to **AF0**, ranging from 1.041 to 1.0705. Sample **AF2** displayed slightly lower O.D. values compared to **AF0** and **AF1**, indicating moderate biocompatibility. Sample **AF3** showed the lowest viability, with O.D. values lower than the control and other samples, particularly at higher concentrations (0.9085 ± 0.0145 at 10 mg/mL), though still indicating overall non-toxicity. These results suggest that all the synthesized hydrogels are nontoxic and showed good biocompatibility.

The live/dead assay results for the samples after 48 h (**AF3**, **AF2**, **AF1**, **AF0**) demonstrate predominantly green fluorescence, indicating high cell viability as shown in Fig. 7. Sample **AF3**, displays almost entirely green fluorescence with few red cells (dead cells), indicating a high percentage of live cells and minimal cytotoxicity. Sample **AF2**, shows a similar pattern with a predominance of green fluorescence and very few red cells, suggesting high cell viability and confirming that the treatment applied to sample **AF2** does not induce significant cytotoxic effects. Sample **AF1**, also exhibits mostly green fluorescence with few red cells, indicating a high viability rate. The fluorescence pattern is consistent with samples **AF3** and **AF2**, confirming the non-toxic nature of the hydrogels. Sample **AF0** also shows predominantly green fluorescence with minimal red staining, further indicating high cell viability and low cytotoxicity. Overall, the live/dead assay results indicate that all the synthesized hydrogels support high levels of cell viability with minimal cytotoxicity, as evidenced by the predominance of green fluorescence and the low incidence of red-stained cells, suggesting that the hydrogels are non-toxic and conducive to maintaining cell health.

4. Conclusions

In a nutshell, we have successfully developed a 3D-printable amyloid nanofibril-based hydrogel encapsulating curcumin, designed for potential drug delivery and hemostatic applications. This proposed hydrogel

was supposedly effective for the delivery of hydrophobic drugs (curcumin) which is normally found difficult to be incorporated in aqueous systems. Amyloid nanofibrils from BSA provided mechanical integrity to the hydrogel and the fibril-drug interaction facilitated increased solubility and encapsulation of curcumin in the hydrogel system while modulating the rheology of the hydrogel through secondary interactions. The printability and rheological studies show the effect of curcumin interaction with the fibril on the hydrogel network and rheology. Also, DPPH and ABTS assays exhibits enhanced free radical scavenging activity for the hydrogels with curcumin compared to the one without curcumin which will also be advantageous for wound healing applications. Among the prepared hydrogel formulations, **AF2** and **AF3** showed acceptable printability with a printability factor close to 1 and post-printing stability of printed scaffolds, characterized by $G' > G''$ in the range of 1 to 10 rad/s and gradual decrease in viscosity with increasing shear rate. **AF2** showed enhanced swelling percentage around 550% and micrographs obtained from SEM also indicates the porous structure of the printed scaffolds favourable for the cell proliferation and nutrient and oxygen diffusion under ambient conditions. The sustained release of curcumin as observed in amyloid nanofibril-based hydrogel makes it a suitable matrix for effective drug delivery. **AF2** and **AF3** also showed hemocompatibility and excellent hemostatic properties, and the coagulation ability of the hydrogel mainly occurs through intrinsic pathway of thrombosis. Furthermore, the in vitro cytotoxicity studies confirmed the hydrogel's biocompatibility to fibroblast cells (L929). Unlike many previously reported amyloid fibril-based hydrogels, the developed hydrogel demonstrates systematic and optimized 3D printability with promising hemostatic property and antioxidant activity in a single system. Formulations **AF2** and **AF3** showed promising structural fidelity and precision, allowing customized scaffold fabrication. Overall, these findings suggest that curcumin-loaded amyloid-based hydrogel could have great potential for use in regenerative

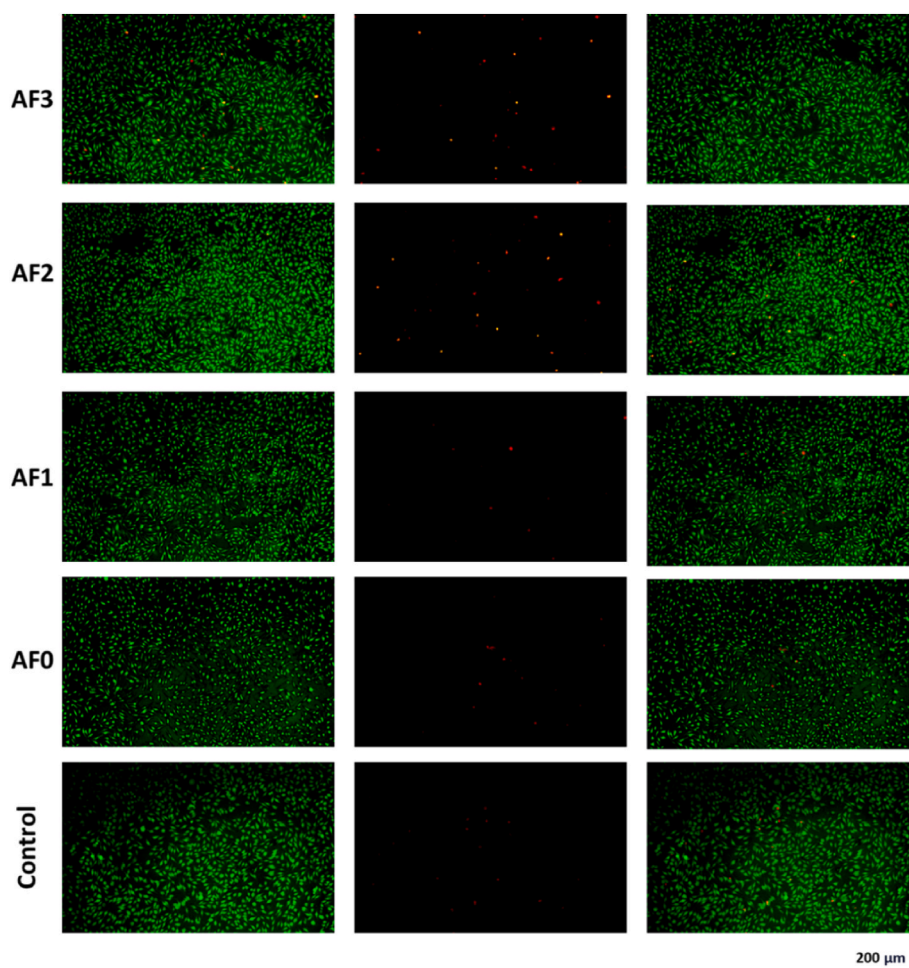


Fig. 7. Live/dead staining of L929 cells after treatment with different hydrogels at (10 mg/mL) after 48 h.

medicine and drug delivery applications.

5. Future prospects

Although the hydrogel exhibited excellent cytocompatibility, hemocompatibility, antioxidant activity, and hemostatic behavior, the biological evaluation was limited to in vitro assays, and in vivo wound healing and biodegradation studies are necessary to validate its therapeutic efficacy and safety under physiological conditions. Also, the long-term mechanical stability and degradation kinetics of the printed scaffolds under dynamic biological environments were not explored. The biological study is limited to a single fibroblast cell line which can be useful for the preliminary cytocompatibility assessment but is insufficient for broader therapeutic claims. In future work, focus will be given on biological evaluation with multiple relevant cell lines, mechanistic investigations, in vivo evaluation of wound healing performance, detailed studies on scaffold degradation kinetics and remodeling, and investigation of cell–material interactions using advanced skin-relevant models and cell lines. Also, further optimization of fibril–drug interactions and printing parameters for patient-specific scaffold designs with controlled drug release profiles with the help of 3D printing will also be on the top priority. We believe that the current research work will pave the way in the near future to design novel 3D-printable amyloid-based hydrogels for applications in regenerative medicine and localized therapeutic delivery.

CRediT authorship contribution statement

D.A. Gouripriya: Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Prama Adhya:** Investigation, Writing – original draft. **Arup Paul:** Formal analysis, Investigation. **Jaideep Adhikari:** Writing – review & editing. **Pratik Das:** Investigation, Writing – original draft. **Agnibin Kundu:** Investigation, Writing – original draft. **Aditya Dev Rajora:** Writing – review & editing. **Piyali Basak:** Supervision, Validation. **Sabu Thomas:** Supervision, Validation. **Prosenjit Saha:** Supervision, Validation, Writing – review & editing. **Pooja Ghosh:** Conceptualization, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

PG is grateful to WB-DSTBT, Government of West Bengal (File No. STBT-11012(25)/12/2024-ST SEC) for financial support. PS acknowledges the funding assistance from SERB-CRG (CRG/2021/005113) and Gouripriya D A acknowledges the DST- INSPIRE (IF220058), Ministry of Science and Technology, Government of India for her fellowship. The authors would like to thank MG University Kottayam for providing access to the FTIR facility. The authors would also like to thank Professor

Manish Kaushal, Department of Chemical Engineering, IIT Kharagpur for helping in rheological measurements.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jddst.2026.108262>.

Data availability

Data will be made available on request.

References

- [1] S.S. Yadav, P.K. Padhy, A.K. Singh, S. Sharma, Tanu, S. Fatima, A. Sinha, R. Tariq, Varsha, S.K. Sharma, S. Priya, Advancements in amyloid-based biological materials for healthcare, environmental and sensing applications, *Mater. Adv.* 5 (2024) 4078–4090, <https://doi.org/10.1039/D3MA00969F>.
- [2] P. Ghosh, P. De, Modulation of amyloid protein fibrillation by synthetic polymers: recent advances in the context of neurodegenerative diseases, *ACS Appl. Bio Mater.* 3 (2020) 6598–6625, <https://doi.org/10.1021/acsabm.0c01021>.
- [3] C.-Y. Chang, T.-L. Hsu, Y.-R. Lai, J.W. Wu, S.S.-S. Wang, S.-C. How, T.-H. Lin, The application of poly (acrylic Acid)-whey protein isolate amyloid Fibril-derived hybrid hydrogels in drug delivery, *J. Drug Deliv. Sci. Technol.* 113 (2025) 107299, <https://doi.org/10.1016/j.jddst.2025.107299>.
- [4] P. Ghosh, A. Kundu, D. Ganguly, From experimental studies to computational approaches: recent trends in designing novel therapeutics for amyloidogenesis, *J. Mater. Chem. B* 13 (2025) 858–881, <https://doi.org/10.1039/D4TB01890G>.
- [5] P. Ghosh, A. Mondal, P. De, Polymeric materials as a promising platform for suppressing protein aggregation process, *Curr. Indian Sci.* 02 (2024), <https://doi.org/10.2174/012210299X290674240509111832>.
- [6] C.F. Monteiro, M.C. Gomes, P. Bharmoria, M.G. Freire, J.A.P. Coutinho, C. A. Custódio, J.F. Mano, Human platelet lysate-derived nanofibrils as building blocks to produce free-standing membranes for cell self-aggregation, *ACS Nano* 18 (2024) 15815–15830, <https://doi.org/10.1021/acsnano.4c02790>.
- [7] N.C. Gomes-da-Silva, M.A. Gomes Soares, E. Ricci-Junior, M. do S. do N. Amorim, L.M.R. Alencar, P. Bhattarai, L. Zhu, C. Follmer, R. Santos-Oliveira, Nanomicelles of taurine inhibit the fibrillation and toxicity of amyloid β 1-42 peptide, *J. Drug Deliv. Sci. Tech.* 86 (2023) 104721, <https://doi.org/10.1016/j.jddst.2023.104721>.
- [8] M. Alrouji, H.M. Al-Kuraishy, A.I. Al-Gareeb, A. Alexiou, M. Papadakis, H.M. Saad, G.E.-S. Batiha, The potential role of human islet amyloid polypeptide in type 2 diabetes mellitus and Alzheimer's diseases, *Diabetol. Metab. Syndr.* 15 (2023) 101, <https://doi.org/10.1186/s13098-023-01082-1>.
- [9] V. Kumar, N. Sinha, A.K. Thakur, Necessity of regulatory guidelines for the development of amyloid based biomaterials, *Biomater. Sci.* 9 (2021) 4410–4422, <https://doi.org/10.1039/D1BM00059D>.
- [10] G. Wei, Z. Su, N.P. Reynolds, P. Arosio, I.W. Hamley, E. Gazit, R. Mezzenga, Self-assembling peptide and protein amyloids: from structure to tailored function in nanotechnology, *Chem. Soc. Rev.* 46 (2017) 4661–4708, <https://doi.org/10.1039/C6CS00542J>.
- [11] T.P.J. Knowles, R. Mezzenga, Amyloid fibrils as building blocks for natural and artificial functional materials, *Adv. Mater.* 28 (2016) 6546–6561, <https://doi.org/10.1002/adma.201505961>.
- [12] J. Zhao, P. Yang, Amyloid-mediated fabrication of organic-inorganic hybrid materials and their biomedical applications, *Adv. Mater. Interfac.* 7 (2020), <https://doi.org/10.1002/admi.202001060>.
- [13] G. Shan, Z. Xu, L. Jiang, Y. Zhang, X. Sui, Fabrication and characterization of glycerol-plasticized soy protein amyloid fibril scaffolds by unidirectional freeze casting method, *Food Hydrocoll.* 147 (2024) 109400, <https://doi.org/10.1016/j.foodhyd.2023.109400>.
- [14] S. Das, R.S. Jacob, K. Patel, N. Singh, S.K. Maji, Amyloid fibrils: versatile biomaterials for cell adhesion and tissue engineering applications, *Biomacromolecules* 19 (2018) 1826–1839, <https://doi.org/10.1021/acs.biomac.8b00279>.
- [15] W. Wang, R.A. Azizyan, A. Garro, A.V. Kajava, S. Ventura, Multifunctional amyloid oligomeric nanoparticles for specific cell targeting and drug delivery, *Biomacromolecules* 21 (2020) 4302–4312, <https://doi.org/10.1021/acs.biomac.0c01103>.
- [16] C. Li, R. Qin, R. Liu, S. Miao, P. Yang, Functional amyloid materials at surfaces/interfaces, *Biomater. Sci.* 6 (2018) 462–472, <https://doi.org/10.1039/C7BM01124E>.
- [17] J. Li, F. Zhang, Amyloids as building blocks for macroscopic functional materials: designs, applications and challenges, *Int. J. Mol. Sci.* 22 (2021) 10698, <https://doi.org/10.3390/ijms221910698>.
- [18] X. Qi, Y. Li, Y. Xiang, Y. Chen, Y. Shi, X. Ge, B. Zeng, J. Shen, Hyperthermia-enhanced immunoregulation hydrogel for oxygenation and ROS neutralization in diabetic foot ulcers, *Cell Biomater* 1 (2025) 100020, <https://doi.org/10.1016/j.celbio.2025.100020>.
- [19] P. Seth, A. Mukherjee, N. Sarkar, Formation of hen egg white lysozyme derived amyloid-based hydrogels using different gelation agents: a potential tool for drug delivery, *Int. J. Biol. Macromol.* 253 (2023) 127177, <https://doi.org/10.1016/j.ijbiomac.2023.127177>.
- [20] N. Singh, K. Patel, A. Navalkar, P. Kadu, D. Datta, D. Chatterjee, S. Mukherjee, R. Shaw, N. Gahlot, A. Shaw, S. Jadhav, S.K. Maji, Amyloid fibril-based thixotropic hydrogels for modeling of tumor spheroids in vitro, *Biomaterials* 295 (2023) 122032, <https://doi.org/10.1016/j.biomaterials.2023.122032>.
- [21] O. Kammona, E. Tsanaksidou, C. Kiparissides, Recent developments in 3d-(bio) printed hydrogels as wound dressings, *Gels* 10 (2024) 147, <https://doi.org/10.3390/gels10020147>.
- [22] S. Cheng, H. Wang, X. Pan, C. Zhang, K. Zhang, Z. Chen, W. Dong, A. Xie, X. Qi, Dendritic hydrogels with robust inherent antibacterial properties for promoting bacteria-infected wound healing, *ACS Appl. Mater. Interfaces* 14 (2022) 11144–11155, <https://doi.org/10.1021/acsami.1c25014>.
- [23] Y. Xiang, P. Zhuge, X. Qi, X. Ge, J. Xiang, H. Xu, E. Cai, Y. Lan, X. Chen, Y. Li, Y. Shi, J. Shen, J. Liu, A cuttlefish ink nanoparticle-reinforced biopolymer hydrogel with robust adhesive and immunomodulatory features for treating oral ulcers in diabetes, *Bioact. Mater.* 39 (2024) 562–581, <https://doi.org/10.1016/j.bioactmat.2024.04.022>.
- [24] Y. Lu, H. Zhao, X. Zhang, B. Qi, Y. Li, Construction of self-assembling hydrogels by interaction of soy protein nanofibrils with polyphenols: insights from intermolecular interaction and gel properties, *Food Biosci.* 69 (2025) 106843, <https://doi.org/10.1016/j.fbio.2025.106843>.
- [25] K.E. Marshall, K.L. Morris, D. Charlton, N. O'Reilly, L. Lewis, H. Walden, L. C. Serpell, Hydrophobic, aromatic, and electrostatic interactions play a central role in amyloid fibril formation and stability, *Biochemistry* 50 (2011) 2061–2071, <https://doi.org/10.1021/bi101936c>.
- [26] H. Khalesi, C. Sun, J. He, W. Lu, Y. Fang, The role of amyloid fibrils in the modification of whey protein isolate gels with the form of stranded and particulate microstructures, *Food Res. Int.* 140 (2021) 109856, <https://doi.org/10.1016/j.foodres.2020.109856>.
- [27] Y. Hu, J.-O. You, J. Aizenberg, Micropatterned hydrogel surface with high-aspect-ratio features for cell guidance and tissue growth, *ACS Appl. Mater. Interfaces* 8 (2016) 21939–21945, <https://doi.org/10.1021/acsami.5b12268>.
- [28] E.Y. Chi, S.L. Frey, A. Winans, K.L.H. Lam, K. Kjaer, J. Majewski, K.Y.C. Lee, Amyloid- β fibrillogenesis seeded by interface-induced peptide misfolding and self-assembly, *Biophys. J.* 98 (2010) 2299–2308, <https://doi.org/10.1016/j.bpj.2010.01.056>.
- [29] M. Usueli, T. Germerdonk, Y. Cao, M. Peydayesh, M. Bagnani, S. Handschin, G. Nyström, R. Mezzenga, Polysaccharide-reinforced amyloid fibril hydrogels and aerogels, *Nanoscale* 13 (2021) 12534–12545, <https://doi.org/10.1039/D1NR03133C>.
- [30] J. Soroia, W. Yanen, Q. Wei, K. Zhang, T. Lu, B. Zhang, A review on biocompatibility nature of hydrogels with 3D printing techniques, tissue engineering application and its future prospective, *Bio-Des. Manufact.* 1 (2018) 265–279, <https://doi.org/10.1007/s42242-018-0029-7>.
- [31] N. Singh, K. Patel, A. Navalkar, P. Kadu, D. Datta, D. Chatterjee, A. Shaw, S. Jadhav, S.K. Maji, Amyloid fibril-based hydrogels for high-throughput tumor spheroid modeling, *bioRxiv* (2020), <https://doi.org/10.1101/2020.12.28.424634>.
- [32] K. Naik, P. Singh, M. Yadav, S.K. Srivastava, S. Tripathi, R. Ranjan, P. Dhar, A. K. Verma, S. Chaudhary, A.S. Parmar, 3D printable, injectable amyloid-based composite hydrogel of bovine serum albumin and aloe vera for rapid diabetic wound healing, *J. Mater. Chem. B* 11 (2023) 8142–8158, <https://doi.org/10.1039/D3TB01151H>.
- [33] Q. Xuan, F. Jiang, H. Dong, W. Zhang, F. Zhang, T. Ma, J. Zhuang, J. Yu, Y. Wang, H. Shen, C. Chen, P. Wang, Bioinspired intrinsic versatile hydrogel fabricated by amyloid toxin simulant-based nanofibrous assemblies for accelerated diabetic wound healing, *Adv. Funct. Mater.* 31 (2021), <https://doi.org/10.1002/adfm.202106705>.
- [34] Y. Ha, Y. Kwon, E.-J. Nam, H. Park, S.R. Paik, Disulfide-mediated elongation of amyloid fibrils of α -synuclein for use in producing self-healing hydrogel and dye-absorbing aerogel, *Acta Biomater.* 145 (2022) 52–61, <https://doi.org/10.1016/j.actbio.2022.04.012>.
- [35] Q. Yang, Y. Miao, J. Luo, Y. Chen, Y. Wang, Amyloid fibril and clay nanosheet dual-nanoengineered dna dynamic hydrogel for vascularized bone regeneration, *ACS Nano* 17 (2023) 17131–17147, <https://doi.org/10.1021/acsnano.3c04816>.
- [36] B.A. Vernaglia, J. Huang, E.D. Clark, Guanidine hydrochloride can induce amyloid fibril formation from hen egg-white lysozyme, *Biomacromolecules* 5 (2004) 1362–1370, <https://doi.org/10.1021/bm0498979>.
- [37] Z. Feng, Y. Li, Y. Bai, Elevated temperatures accelerate the formation of toxic amyloid fibrils of hen egg-white lysozyme, *Vet. Med. Sci.* 7 (2021) 1938–1947, <https://doi.org/10.1002/vms3.522>.
- [38] S.-Y. Ow, D.E. Dunstan, The effect of concentration, temperature and stirring on hen egg white lysozyme amyloid formation, *Soft Matter* 9 (2013) 9692, <https://doi.org/10.1039/c3sm51671g>.
- [39] S. Almoahammed, M. Alruwaili, E.G. Reynaud, G. Redmond, J.H. Rice, B. J. Rodriguez, 3D-printed peptide-hydrogel nanoparticle composites for surface-enhanced raman spectroscopy sensing, *ACS Appl. Nano Mater.* 2 (2019) 5029–5034, <https://doi.org/10.1021/acsnanm.9b00940>.
- [40] P. Lou, S. Liu, Y. Wang, C. Pan, X. Xu, M. Zhao, G. Liao, G. Yang, Y. Yuan, L. Li, J. Zhang, Y. Chen, J. Cheng, Y. Lu, J. Liu, Injectable self-assembling peptide nanofiber hydrogel as a bioactive 3D platform to promote chronic wound tissue regeneration, *Acta Biomater.* 135 (2021) 100–112, <https://doi.org/10.1016/j.actbio.2021.08.008>.
- [41] Q. Li, G. Qi, D. Lutter, W. Beard, C.R.S. Souza, M.A. Highland, W. Wu, P. Li, Y. Zhang, A. Atala, X. Sun, Injectable peptide hydrogel encapsulation of mesenchymal stem cells improved viability, stemness, anti-inflammatory effects, and early stage wound healing, *Biomolecules* 12 (2022) 1317, <https://doi.org/10.3390/biom12091317>.

- [42] A.C. Farsheed, A.J. Thomas, B.H. Pogostin, J.D. Hartgerink, 3D printing of self-assembling nanofibrous multidomain peptide hydrogels, *Adv. Mater.* 35 (2023), <https://doi.org/10.1002/adma.202210378>.
- [43] G. D. A., J. Adhikari, P. Debnath, S. Ghosh, P. Ghosh, S. Thomas, E. Ghandilyan, P. Gorbato, E. Kuchukyan, S. Gasparyan, P. Saha, 3D printing of bacterial cellulose for potential wound healing applications: current trends and prospects, *Int. J. Biol. Macromol.* 279 (2024) 135213, <https://doi.org/10.1016/j.ijbiomac.2024.135213>.
- [44] A. Singh, P. Ghosh, S. Mukherjee, A.K. Ojha, A. Hansda, P. Choudhury, S. Halder, S. Sharma, G. Mukherjee, S. Dasgupta, K. Chaudhury, Transition metallo-curcumin complexes: a new hope for endometriosis? *J. Mater. Chem. B* 10 (2022) 9682–9698, <https://doi.org/10.1039/D2TB01651F>.
- [45] F. Ahmadi, J. Akbari, M. Saeedi, M. Seyedabadi, P. Ebrahimnejad, S. Ghasemi, A. Nokhodchi, Efficient synergistic combination effect of curcumin with piperine by polymeric magnetic nanoparticles for breast cancer treatment, *J. Drug Deliv. Sci. Tech.* 86 (2023) 104624, <https://doi.org/10.1016/j.jddst.2023.104624>.
- [46] M. Rajabzadeh-Khosroshahi, M. Pourmadadi, F. Yazdian, H. Rashedi, M. Navaei-Nigjeh, B. Rasekh, Chitosan/agarose/graphitic carbon nitride nanocomposite as an efficient pH-sensitive drug delivery system for anticancer curcumin releasing, *J. Drug Deliv. Sci. Technol.* 74 (2022) 103443, <https://doi.org/10.1016/j.jddst.2022.103443>.
- [47] K. Mahmood, K.M. Zia, M. Zuber, M. Salman, M.N. Anjum, Recent developments in curcumin and curcumin based polymeric materials for biomedical applications: a review, *Int. J. Biol. Macromol.* 81 (2015) 877–890, <https://doi.org/10.1016/j.ijbiomac.2015.09.026>.
- [48] S. Omid, Z. Rafiee, A. Kakanejadifard, Design and synthesis of curcumin nanostructures: evaluation of solubility, stability, antibacterial and antioxidant activities, *Bioorg. Chem.* 116 (2021) 105308, <https://doi.org/10.1016/j.bioorg.2021.105308>.
- [49] R. Pan, G. Liu, Y. Zeng, X. He, Z. Ma, Y. Wei, S. Chen, L. Yang, L. Tao, A multi-responsive self-healing hydrogel for controlled release of curcumin, *Polym. Chem.* 12 (2021) 2457–2463, <https://doi.org/10.1039/D1PY00176K>.
- [50] M.M. El Taweel, M.A. Tawfik, K. Soliman, M.S. Khattab, M.M. Farag, Tailoring of topically applied curcumin loaded pro-nanovesicles for skin cancer treatment: In-vitro characterization, statistical optimization and histopathological assessment of subcutaneous Ehrlich carcinoma mice model, *J. Drug Deliv. Sci. Technol.* 88 (2023) 104957, <https://doi.org/10.1016/j.jddst.2023.104957>.
- [51] K. Pan, Y. Luo, Y. Gan, S.J. Baek, Q. Zhong, pH-driven encapsulation of curcumin in self-assembled casein nanoparticles for enhanced dispersibility and bioactivity, *Soft Matter* 10 (2014) 6820, <https://doi.org/10.1039/C4SM00239C>.
- [52] D. Yuan, F. Zhou, P. Shen, Y. Zhang, L. Lin, M. Zhao, Self-assembled soy protein nanoparticles by partial enzymatic hydrolysis for pH-driven encapsulation and delivery of hydrophobic cargo curcumin, *Food Hydrocoll.* 120 (2021) 106759, <https://doi.org/10.1016/j.foodhyd.2021.106759>.
- [53] L. Pathak, A. Kanwal, Y. Agrawal, Curcumin loaded self assembled lipid-biopolymer nanoparticles for functional food applications, *J. Food Sci. Technol.* 52 (2015) 6143–6156, <https://doi.org/10.1007/s13197-015-1742-2>.
- [54] M.N. Oskouie, N.S. Aghili Moghaddam, A.E. Butler, P. Zamani, A. Sahebkar, Therapeutic use of curcumin-encapsulated and curcumin-primed exosomes, *J. Cell. Physiol.* 234 (2019) 8182–8191, <https://doi.org/10.1002/jcp.27615>.
- [55] F. Aqil, R. Munagala, J. Jeyabalan, A.K. Agrawal, R. Gupta, Exosomes for the enhanced tissue bioavailability and efficacy of curcumin, *AAPS J.* 19 (2017) 1691–1702, <https://doi.org/10.1208/s12248-017-0154-9>.
- [56] D. Sun, X. Zhuang, X. Xiang, Y. Liu, S. Zhang, C. Liu, S. Barnes, W. Grizzle, D. Miller, H.-G. Zhang, A novel nanoparticle drug delivery system: the anti-inflammatory activity of curcumin is enhanced when encapsulated in exosomes, *Mol. Ther.* 18 (2010) 1606–1614, <https://doi.org/10.1038/mt.2010.105>.
- [57] Y.-Q. Xu, W.-R. Chen, J.K. Tsosie, X. Xie, P. Li, J.-B. Wan, C.-W. He, M.-W. Chen, Niosome encapsulation of curcumin: characterization and cytotoxic effect on ovarian cancer cells, *J. Nanomater.* 2016 (2016) 1–9, <https://doi.org/10.1155/2016/6365295>.
- [58] Z. Sadeghi Ghadi, P. Ebrahimnejad, Curcumin entrapped hyaluronan containing niosomes: preparation, characterisation and in vitro/in vivo evaluation, *J. Microencapsul.* 36 (2019) 169–179, <https://doi.org/10.1080/02652048.2019.1617360>.
- [59] Z. Zhang, R. Zhang, L. Zou, L. Chen, Y. Ahmed, W. Al Bishri, K. Balamash, D. J. McClements, Encapsulation of curcumin in polysaccharide-based hydrogel beads: impact of bead type on lipid digestion and curcumin bioaccessibility, *Food Hydrocoll.* 58 (2016) 160–170, <https://doi.org/10.1016/j.foodhyd.2016.02.036>.
- [60] J.M. Silva, E. Silva, R.L. Reis, Therapeutic deep eutectic solvents assisted the encapsulation of curcumin in alginate-chitosan hydrogel beads, *Sustain. Chem. Pharm.* 24 (2021) 100553, <https://doi.org/10.1016/j.scp.2021.100553>.
- [61] J. Liu, Z. Chen, J. Wang, R. Li, T. Li, M. Chang, F. Yan, Y. Wang, Encapsulation of curcumin nanoparticles with MMP9-responsive and thermos-sensitive hydrogel improves diabetic wound healing, *ACS Appl. Mater. Interfaces* 10 (2018) 16315–16326, <https://doi.org/10.1021/acsami.8b03868>.
- [62] Y. Fan, C. Gan, H. Zhang, J. Yi, Characteristics, physicochemical stability and in vitro release of curcumin-loaded glycated bovine serum albumin nanofibrils: effects of molecular weight of saccharide, *Food Hydrocoll.* 155 (2024) 110210, <https://doi.org/10.1016/j.foodhyd.2024.110210>.
- [63] K. Nayak, P. Ghosh, S. Barman, B. Sudhamalla, P. Theato, P. De, Amyloid β -peptide segment conjugated side-chain proline-based polymers as potent inhibitors in lysozyme amyloidosis, *Bioconj. Chem.* 35 (2024) 312–323, <https://doi.org/10.1021/acs.bioconjchem.3c00509>.
- [64] A. Bera, P. Ghosh, K. Goswami, P. De, Amino acid-based polymer-coated silver nanoparticles as insulin fibril inhibitors, *ACS Appl. Nano Mater.* 6 (2023) 8705–8716, <https://doi.org/10.1021/acsnm.3c01078>.
- [65] F. Zehtabi, A. Gangrade, K. Tseng, R. Haghniaz, R. Abbasgholizadeh, H. Montazerian, D. Khorsandi, J. Bahari, A. Ahari, N. Mohaghegh, N. Hosseinzadeh Kouchehbaghi, K. Mandal, M. Mecwan, A. Rashad, N. Roberto de Barros, Y. Byun, M. Ermis, H. Kim, A. Khademhosseini, Injectable shear-thinning hydrogels with sclerosing and matrix metalloproteinase modulatory properties for the treatment of vascular malformations, *Adv. Funct. Mater.* 33 (2023), <https://doi.org/10.1002/adfm.202305880>.
- [66] R. Ramakrishnan, N. Kasoju, R. Raju, R. Geevarghese, A. Gauthaman, A. Bhatt, Exploring the potential of alginate-gelatin-diethylaminoethyl cellulose-fibrinogen based bioink for 3D bioprinting of skin tissue constructs, *Carbohydr. Polym. Technol. Appl.* 3 (2022) 100184, <https://doi.org/10.1016/j.carpta.2022.100184>.
- [67] P. Ghosh, A. Singha Roy, S. Chaudhury, S.K. Jana, K. Chaudhury, S. Dasgupta, Preparation of albumin based nanoparticles for delivery of fisetin and evaluation of its cytotoxic activity, *Int. J. Biol. Macromol.* 86 (2016) 408–417, <https://doi.org/10.1016/j.ijbiomac.2016.01.082>.
- [68] F. Hozan, E.N. Uslu, G.Ş. Karatoprak, Ç. Yücel, Baicalein-loaded aspasomal formulations: development, characterization and evaluation of antioxidant and anti-inflammatory effects, *J. Pharm. Innov.* 20 (2025) 71, <https://doi.org/10.1007/s12247-025-09979-2>.
- [69] W. Deng, Y. Liu, C. He, X. Xiong, R. Zhang, T. Yan, S. Shi, D.-G. Yu, H. Yang, Synergistic improvements of properties of cellulose acetate based curcumin@TiO₂ nanofibers via triaxial electrospraying, *Chem. Eng. J.* 506 (2025) 160117, <https://doi.org/10.1016/j.cej.2025.160117>.
- [70] P. Snetkov, E. Rogacheva, A. Kremleva, S. Morozkina, M. Uspenskaya, L. Kraeva, In-vitro antibacterial activity of curcumin-loaded nanofibers based on hyaluronic acid against multidrug-resistant ESKAPE pathogens, *Pharmaceutics* 14 (2022) 1186, <https://doi.org/10.3390/pharmaceutics14061186>.
- [71] P. Ghosh, S. Bag, S. Parveen, E. Subramani, K. Chaudhury, S. Dasgupta, Nanoencapsulation as a promising platform for the delivery of the morin-cu(ii) complex: antibacterial and anticancer potential, *ACS Omega* 7 (2022) 7931–7944, <https://doi.org/10.1021/acsomega.1c06956>.
- [72] R.K. Sen, P. Prabhakar, Shruti, P. Verma, A. Vikram, A. Mishra, A. Dwivedi, V. S. Gowri, J.P. Chaurasia, D.P. Mondal, A.K. Srivastava, N. Dwivedi, C. Dhand, Smart nanofibrous hydrogel wound dressings for dynamic infection diagnosis and control: soft but functionally rigid, *ACS Appl. Bio Mater.* 7 (2024) 999–1016, <https://doi.org/10.1021/acsabm.3c01000>.
- [73] P. Ghosh, S. Bag, A. Singha Roy, E. Subramani, K. Chaudhury, S. Dasgupta, Solubility enhancement of morin and epicatechin through encapsulation in an albumin based nanoparticle system and their anticancer activity against the MDA-MB-468 breast cancer cell line, *RSC Adv.* 6 (2016) 101415–101429, <https://doi.org/10.1039/C6RA20441D>.
- [74] A. Bera, D. Mukhopadhyay, K. Goswami, P. Ghosh, R. De, P. De, Fatty acid-based polymeric micelles to ameliorate amyloidogenic disorders, *Biomater. Sci.* 10 (2022) 3466–3479, <https://doi.org/10.1039/D2BM00359G>.
- [75] O. Holovko, O. Dmytrenko, M. Kulish, A. Lesiuk, O. Pavlenko, A. Naumenko, M. Kaniuk, I. Doroshenko, Mechanisms of heteroassociation in aqueous solutions of BSA with curcumin, *J. Mol. Liq.* 415 (2024) 126364, <https://doi.org/10.1016/j.molliq.2024.126364>.
- [76] A. Bera, P. Ghosh, K. Goswami, P. De, Amino acid-based polymer-coated silver nanoparticles as insulin fibril inhibitors, *ACS Appl. Nano Mater.* 6 (2023) 8705–8716, <https://doi.org/10.1021/acsnm.3c01078>.
- [77] H. Ramshini, M. Mohammad-zadeh, A. Ebrahim-Habibi, Inhibition of amyloid fibril formation and cytotoxicity by a chemical analog of Curcumin as a stable inhibitor, *Int. J. Biol. Macromol.* 78 (2015) 396–404, <https://doi.org/10.1016/j.ijbiomac.2015.04.038>.
- [78] A. Krishnamoorthy, N. Tavoosi, G.K.L. Chan, J. Liu, G. Ren, G. Cavignoli, R. O. Ryan, Effect of curcumin on amyloid-like aggregates generated from methionine-oxidized apolipoprotein A-I, *FEBS Open Bio* 8 (2018) 302–310, <https://doi.org/10.1002/2211-5463.12372>.
- [79] B. Khurshid, A.U. Rehman, S. Muhammad, A. Wadood, J. Anwar, Toward the noninvasive diagnosis of alzheimer's disease: molecular basis for the specificity of curcumin for fibrillar amyloid- β , *ACS Omega* 7 (2022) 22032–22038, <https://doi.org/10.1021/acsomega.2c02995>.
- [80] T. Zhang, W. Zhao, Z. Xiahou, X. Wang, K. Zhang, J. Yin, Bioink design for extrusion-based bioprinting, *Appl. Mater. Today* 25 (2021) 101227, <https://doi.org/10.1016/j.apmt.2021.101227>.
- [81] S. Tian, H. Zhao, N. Lewinski, Key parameters and applications of extrusion-based bioprinting, *Bioprinting* 23 (2021) e00156, <https://doi.org/10.1016/j.bprint.2021.e00156>.
- [82] A. Chalard, P. Joseph, S. Souleille, B. Lonetti, N. Saffon-Merceron, I. Loubinoux, L. Vaysse, L. Malaquin, J. Fitremann, Wet spinning and radial self-assembly of a carbohydrate low molecular weight gelator into well organized hydrogel filaments, *Nanoscale* 11 (2019) 15043–15056, <https://doi.org/10.1039/C9NR02727K>.
- [83] Y. Boulmouk, K. Belguidoum, F. Meddour, H. Amira-Guebailia, Enhanced antioxidant properties of novel curcumin derivatives: a comprehensive DFT computational study, *Struct. Chem.* 35 (2024) 825–839, <https://doi.org/10.1007/s11224-023-02237-6>.
- [84] T. Li, D. Wang, X. Zhang, L. Wang, Structural characterization and binding interaction of rice glutelin fibrils complexing with curcumin, *Food Chem.* 448 (2024) 139063, <https://doi.org/10.1016/j.foodchem.2024.139063>.